Distributional Learning as Grammaticalization: Modals in the History of English

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July 2024

Abstract

In this paper, we approach the phenomenon of grammaticalization from the perspective of child language. Building on recent findings in infant language research and learnability studies, we propose that syntactic categories are distributionally constructed by children: the classification of words into syntactic categories may change in response to changes in the linguistic data over time, resulting resulting in grammaticalization. Using the modal verbs in the history of English as a case study, we describe a distributional learning model that invokes the Tolerance Principle (Yang 2016), and its recursive application, as a formal mechanism for syntactic category formation. We show that throughout the history of English, the categorization model can reliably discover two distributional properties, the distribution of negation and inflectional morphology in the syntactic frame, that separate modals from lexical verbs and other auxiliaries. The placement of negation became especially salient after 1650 when the independent loss of verb movement was complete, which helps explain why, among the Germanic language family, only English developed a fully distinctive class of modal verbs. Taken together, our method and results provide fresh impetus for the long-standing research program that places child language acquisition at the center of language change.

Keywords: Language change, grammaticalization, modals, Middle English, Tolerance Principle, distributional learning, child language acquisition

1 introduction

Grammaticalization refers to the well-studied phenomenon where content words become functional words during the course of language change (Meillet 1912). In order to understand grammaticalization, therefore, one must confront fundamental questions about the nature of syntactic categories: What exactly are function words and content words, how is the distinction between them established, and why do they change over time? Integrating research on child language acquisition in the past few decades and especially recent progress on the formal mechanism of learning, we propose that syntactic categories are formal classes of words that children establish on a purely distributional basis. On this view, a critical component of grammaticalization is the child learner's response to the changing distributional properties of the input data that may reflect independent developments in the language: the classification of words may change as a result.

The empirical study of our paper focuses on the development of modals in the history of English (*can*/*could*, *may*/*might*, *shall*/*should*, *will*/*would*, *must*), one of the paradigm case studies of grammaticalization. Our approach builds on a theory of how children categorize words on the basis of distributional data.

This is laid out in Section 2, where we review the developmental literature on very young children's ability to construct formal categories of words. The centerpiece of the theory is the Tolerance Principle (Yang 2005, 2016), a formal mechanism that captures distributional regularities in language. Taking Present-Day English (PDE) as a starting point, we show in Section 3 that children can discover robust distributional evidence to distinguish lexical and functional verbs, including the further distinction between auxiliaries and modals in the latter group. In particular, the distribution of negation and inflectional morphology in positions adjacent to the modal verbs are unique "signatures" that identify them as a unique syntactic category. Importantly, these classes can be established on a very small vocabulary, one which corresponds to the very early stage of child language acquisition. Section 4 is methodological in nature. Despite the very obvious differences between child-directed speech and records of historical language, our distributional learning method is nevertheless applicable to historical data to draw conclusions about the grammars learners must have formed at the time, thanks to the developmental constraints of language acquisition (e.g., very small vocabulary) and the nature of the TP as a model of learning and categorization. In Section 5, we review the previous literature on the development of modals in the history of English, which serves to highlight the formal nature of our approach: modals can be established as an independent syntactic category which is subsequently associated with other syntactic, semantic, and pragmatic properties. Section 6 describes the distributional analyses of Middle English and Early Modern English. While the morphological and syntactic properties of the language changed during these stages of historical development, language learners could nevertheless rely on robust formal distributional data to distinguish modals from other classes of verbs. Section 7 summarizes our findings, including a brief discussion of how our account provides an explanation of the rather unique development of English modals in comparison to related languages. We also consider the nature of our method and its applicability to other cases of language change.

2 the distributional learning of syntactic categories

Syntactic categories are the foundation of grammar but their status in linguistic theory and language acquisition remains an open question. There is considerable debate as to whether syntactic categories are universally attested in the world's languages (Comrie 1989). Even if syntactic categories are indeed innate and/or universal, children still have to learn that, for examples, the English word *cat* is a noun, *see* is a verb, but *jump* may be flexibly used as both. The historical nature of language adds another challenge: as the phenomenon of grammaticalization illustrates, the syntactic category of words is not rigid but can shift over time.

Our approach to grammaticalization follows a formal approach according to which syntactic categories, and indeed all linguistic structures, are distributionally defined. Under this view, syntactic categories are formal classes children create empirically on the basis of the input data. They are not assumed to be innate or universal, and the extent to which they are similar across languages, and among individual speakers within a single language, is entirely due to the similarities in the distributional data and the language acquisition mechanism that is presumably shared across all language learners. Our theory is motivated by two strands of evidence that have emerged in the past two decades of research: (a) infant's sensitivity to formal distributional patterns in language (and other domains), and (b) a general learning mechanism by which formal regularities can be derived from distributional information. We review these findings in turn.

2.1 Distributional regularities in infant language

It is quite clear that the syntactic categories of words often have semantic correlates: in many languages, words denoting objects are used as nouns, actions and events as verbs, properties as adjectives, etc. Modal verbs in English (*can/could*, *may/might*, *shall/should*, *will/would*, *must*), the topic of our empirical study, typically express the meaning of possibility or necessity. According to the semantic bootstrapping hypothesis (Macnamara 1982, Pinker 1984), such correlations are part of the innate endowment for language: learning the meanings of words immediately yield their syntactic categories. Despite its intuitive appeal, the semantic bootstrapping hypothesis is far from a complete solution. As its name entails, the hypothesis is only a start. The meanings of many, perhaps most, words cannot be readily learned from observation – words with abstract meanings such as *seem*, *idea*, *true* – but may rely on the development and use of a formal grammatical system built upon syntactic categories (Gleitman 1990). At the same time, and more critically, an important part of the lexicon consists of functional categories such as auxiliaries, prepositions, determiners, and inflectional markers, which are widely recognized as *lacking* a semantic criterion for classification. Indeed, as we will review in Section 3, English-learning children before age 2 have clearly established modals as a formal category. However, their semantic understanding of these elements takes many more years to develop and thus cannot serve to establish their syntactic status.

Partly for these reasons, the structuralist program in American linguistics (Harris 1951) and generative grammar (Chomsky 1955, 1957) has consistently advocated a formal approach to syntactic categories: words that are distributionally similar are regarded members of the same syntactic category. Recent research on infant's ability for formal and distributional learning of language provides compelling evidence for this approach. In a line of research most prominently developed by Shi and colleagues (e.g., Shi 2023), it is precisely these content-poor functional elements that play the decisive role in syntactic category formation.

Infants at birth are capable of distinguishing functional and lexical words (Shi et al. 1999). They do so presumably based on acoustic grounds: functional words are shorter, less loud, and occupy a smaller and more centralized region of the acoustic space than lexical words (Shi et al. 1998). Before age one, high frequency functional words are accurately represented in the infant's lexicon and are recruited to extract and learn additional words (Shi et al. 2006). Analogously, high frequency inflectional endings are also segmented and recognized by infants (Santelmann & Jusczyk 1998, Mintz 2013). For example, 6-month-old English-learning infants can segment off endings such as *-s*, *-ing*, and *-ed* in novel words; by 8 month, they are able to identify novel stems inflected with *-ing* (Kim & Sundara 2021). Similar findings on the segmentation of inflectional endings and their distributional properties have been reported for a wide range of morphemes and languages (Soderstrom et al. 2007, Höhle et al. 2004, Culbertson et al. 2016). The categorization of words is clearly formal in nature as it can be demonstrated on novel distributional patterns not found in the infant's native language. For example, Marquis & Shi (2012) show that French-learning infants can learn a novel "suffix" (/u/), presented as an ending of a variety of novel words *Fitou*, *Linchou*, *Balou*, *Vaunou*, etc. in a laboratory setting, and carry out distributional analysis akin to back formation in identifying a novel stem *Trid* from an "inflected" form such as /tridu/. Additionally, infants can distinguish grammatical and ungrammatical combinations even across non-local contexts.

It must be pointed out that at this early stage, infant's knowledge of the distributional properties of functional elements is still a long way from the full range of grammatical properties of these elements. It should also be pointed out that none of the distributional patterns reviewed above relies on the semantic aspects of language. Presumably, infants can learn that *-ed* is a segmentable suffix by encountering words such as *walk*, *call*, *want*, *ask*, etc. and their inflected forms *walked*, *called*, *wanted*, and *asked*. They can do so purely formally without needing to know what these words mean or what grammatical function is served by *-ed*. Indeed, infants at this stage of language acquisition understand very few words, perhaps no more than a few dozens (Fenson et al. 1994, Hart & Risley 1995). This initial step of formal learning of segmentation, however, is a prerequisite to developing a fuller understanding of inflectional morphology. In addition, the mere identification of functional elements makes them "anchors" around which syntactic categories can be defined as formal classes.

For example, in a landmark study (Shi & Melançon 2010), 14-month-old French-learning infants hear novel words (e.g., *mige*) following familiar determiners (e.g., *ton* 'your') and *des* ('some'). Afterwards, they expect the novel word to follow other determiners in the language as in *Le mige*, but are surprised if the word follows a pronoun as in *Je mige*. Such findings suggest that infants have formed an equivalent class – which linguists call "determiners" – in which members are expected to share distributional properties. It has also been shown (Babineau et al. 2020), with the same experimental paradigm but under more facilitatory conditions, that infants can also use the preceding subject pronoun such as *Je* and *Il* to identify novel words as verb-like items. Inflectional markers, like words, similarly play a role in syntactic categorization. At 16 months of age, English-learning infants are surprised when listening to the *-ed* suffixed attached to familiar nouns (e.g., *trucked*, *haired*) but not when it is attached to novel items (e.g., /fimd/), which are presumably treated as verb-like items (Figueroa & Gerken 2019).

We make several general observations to conclude this brief review of infant's distributional learning of language. First, the computational mechanisms of tracking predictive dependencies are well supported in both linguistic and non-linguistic domains (Saffran et al. 1996, Marcus et al. 1999). Infants can readily detect salient distributional regularities in artificial language (Gómez 2002) as well as real language for which they have no prior experience (Gerken et al. 2005). Second, while we have focused on purely formal relations – indeed only those defined on linear sequences – for the learning of syntactic categories there is also considerable evidence that other, and domain specific linguistic principles, also guide infant's distributional learning. For example, prosodic boundary has long been established as a constraint on distributional learning of words and syntactic categories (e.g. Mattys et al. 1999, Massicotte-Laforge & Shi 2015). Our point is that formal distributional cues over very simple sequences are *sufficient* for the establishment of a small set of words and their formal syntactic categories: more complex linguistic structures can be constructed subsequently with concomitant development of meaning.

We now turn to study the mechanism by which children construct the syntactic categories from the distributional properties of words.

2.2 The Tolerance Principle as a criterion for syntactic category formation

The identification of syntactic categories has been a major topic in the study of language especially in computational linguistics and natural language processing under the banner of part-of-speech (POS) tagging. The majority of research is on supervised POS tagging where the model has access to an POS-annotated corpus and learns the distributional regularities which are then applied to unannotated data for performance evaluation. Since children obviously do not have similar resources during language acquisition, a more suitable comparison is unsupervised learning for which no data annotation is available. Unsupervised POS tagging remains a highly challenging problem: see Christodoulopoulos et al. (2010) for review. This is not the place to provide a survey of these methods. It is sufficient to point out that these methods, even if they were more successful, would still fall short of the requirements of cognitive models of language acquisition. For example, state-of-the-art unsupervised POS tagging methods (e.g., Stratos et al. 2016) are provided with the number of syntactic categories for the language, which is not available to the child learner. For clustering based approaches (e.g., Redington et al. 1998, Mintz et al. 2002), the size and separation of clusters are controlled by hyper-parameters which are carefully calibrated to achieve the best results but otherwise have no clear psychological motivation. As such, these researchers are explicit in stating that the models are meant to demonstrate the existence/utility of distributional information in the language data, rather than cognitive models of how children may exploit such distributional data. Much like the infant distributional learning studies, these models did confirm that the immediately local contexts of words provide crucial information for syntactic categorization.

In what follows, we describe a new distributional approach to syntactic category formation that does away with the unmotivated assumptions in previous work, and builds on the developmental findings in child language acquisition such as the learning of distributional patterns in infancy just reviewed. Our approach is part of a research program where the goal is to establish a discovery procedure for grammar (Chomsky 1955, 1957): the centerpiece is the Tolerance Principle (Yang 2005, 2016).

The Tolerance Principle (henceforth TP) is a learning principle that specifies the upper-bound of exceptions that a productive pattern can withstand. More specifically, the TP states

> A rule defined over N items can generalize if e, the number of the items that do not follow rules, does not exceed a threshold, $\theta_N = N/\ln N$.

We will not review the motivation or the derivation of the TP; see Yang (2016) for details. The TP is a parameter-free learning model. To generalize or not is determined by two input values (i.e., N and e) which are counts in an individual learner's vocabulary: a categorical prediction is then made without additional degrees of freedom. Because of its simplicity, the TP has been effectively applied to a wide range of learning problems in phonology, morphology, and syntax with implications for language change (e.g., Sneller et al. 2019, Björnsdóttir 2021, van Tuijl & Coopmans 2021, Pearl & Sprouse 2021, Henke 2022, Kodner 2022, Dresher & Lahiri 2022, Ringe & Yang 2022, Trips & Rainsford 2022, Belth 2023, 2024). It has also received support from artificial language studies which manipulate the number of rule-following and rule-defying items to test very precise predictions of the TP (Schuler et al. 2016), including results that support the TP as a domain general learning mechanism (Shi & Emond 2023).

The TP specifies whether a rule is good enough for a given set of words. The failure to do so triggers the subdivision of words along some appropriate dimension such that the TP can be used recursively to establish rules of narrower scopes. This enables children to learn "global" rules where a single productive rule suffices, as in the case of English past tense. It also enables children to learn multiple productive rules defined over subsets of words such as the plethora of plural suffixes in German whose applicability depends on the phonological properties as well as grammatical gender of nouns (Wiese 1996). The recursive application of the TP was explored in depth in Yang (2016); recent work has automated the recursive search for rules in a computational implementation (Belth et al. 2021, 2024).

In the present study, we invoke the TP as a formal principle of category formation. If a set of words is *sufficiently* homogeneous with respect to their distributional properties, then the set can be regarded as members of a single syntactic category: the TP threshold provides a precise value of what counts as "sufficiently" homogeneous. If a class is not sufficiently homogeneous, then subdivision is necessary and additional syntactic categories may be created. The process creates a hierarchy of categories in a decision tree-like fashion; it terminates when all categories are sufficiently homogeneous.

The TP operates over the child's vocabulary which grows incrementally, and very slowly, especially in the beginning stage of language acquisition (Gleitman & Trueswell 2020). This is critical. Many important syntactic categories are functional categories (e.g., prepositions, determiners, modals) that have a closed set of very few members. If situated in a large vocabulary (a large value of N in the TP framework), these items — despite their highly distinctive distributional property — would be tolerated as exceptions to the homogeneity and fail to be established as independent categories. Fortunately, because functional categories are highly frequent and, as reviewed earlier, children learn their distributional properties very early, they will form a major component of the early vocabulary from which their distributional profile will be prominent, reliably triggering the subdivision of words into distinct syntactic categories.

3 modals as a distributional class in present-day english

We now illustrate how the TP operates to create syntactic categories based on the distributional profiles of words in Present-Day English, with specific focus on verbs that include modals.

Consider the ten most frequent phonological words that follow the subject pronoun *I* in a 15 million corpus of child-directed English extracted from the CHILDES database (MacWhinney 2000). We take these as the very first items in a typical child learner's vocabulary, based on the strong evidence that more frequent words are acquired earlier by children (e.g., Goodman et al. 2008):

do, think, will/'ll, am/'m, can, know, want, have, see, got

These words are defined entirely formally and linearly, something that infants are capable of doing as reviewed earlier. Call these "*I*-followers". But even this very small set of words does not form a homogeneous class: their successors on the right side show a sufficiently high level of heterogeneity. In particular, the most frequent successor across all 10 items is the negation *not* and the contracted form *n't*. (In child-directed English, over 90% of negations are contracted.) But out of the 10 items, only the following 5 are robustly attested as "NEG-predecessors":

do, will/'ll, am/m', can, have

For a very small vocabulary of 10 items, the 5 above are in fact sufficient to puncture the distributional homogeneity and trigger a subdivision: the TP threshold θ_{10} is 4. In other words, the original set *must* be subdivided into two contrasting set: *do*, *will/'ll*, *am/'m*, *can* and *have* in one, and *think*, *know*, *want*, *see*, and *got* in the other. These are, of course, what linguists refer to "functional" and "lexical" verbs. For the child learner, they are simply defined as [$\pm$NEG-predecessor], a distributional criterion.

Further division ensues when additional distributional context is taken into account. Focusing on the functional subset, it is immediately clear that only some are robustly followed by bare infinitive verbs: "I can *walk*", "I will *help*", etc. Note that what we refer to as bare infinitives are, to the infant learner, no more than words that alternate with overt inflectional endings such as *-ing* and *-ed*, which they recognize before age 1 and can even learn novel suffixes in a laboratory setting (Marquis & Shi 2012). These items are, of course, exactly the modal verbs of English. Therefore, two distributional signatures — i.e., NEG-preceding and bare-infinitive-taking — are sufficient to define the three major classes of verbs, as illustrated by the decision tree in Figure 1.

Our decision to examine the ten most frequent items may seems arbitrary but is nevertheless indicative of the language acquisition process. Of course not every child will learn exactly these ten as the first items in their vocabulary, but the early vocabulary necessarily contains a high proportion of functional verbs. At some point very early on, probably when the vocabulary size reaches no more than a dozen or so, the functional verbs and the lexical verbs will no longer form a sufficiently homogeneous distributional class à la the TP: subdivision will follow and the creation of the distinct categories is therefore assured. (See Section 4 for discussion of similar scenarios in the historical data.) The modal vs. non-modal subdivision within the functional class is also inevitable. About half of the functional verbs — no more than 20 in all (Zwicky & Pullum 1983) — are modals: TP-based classification will necessarily trigger the modal-vs.-auxiliary split as shown in right branch of the decision tree in Figure 1.

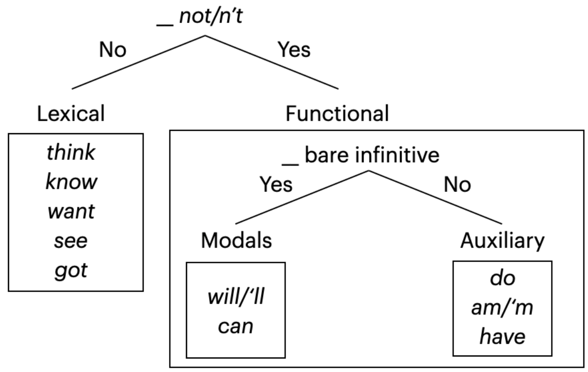

Figure 1: A decision tree that represents the major verbal categories that can be constructed on a very small lexicon.

The decision tree for the categorization of words will continue to grow and be refined. The lexical verb branch will be further subdivided once children learn that a sufficiently large number of them take a complement (e.g., a pronoun such as *it*), thereby creating the intransitive vs. transitive distinction: the variants of *have* and *do* as in "I have it" and "I do that", for instance, will be placed under the appropriate lexical verb branch. It should also be noted that a handful of lexical verbs do take bare infinitives (e.g., *go get pizza*, *come play soccer*, *help carry the bag*). But these will not be classified as modals because they appear before negation. New modals such as *gonna* and *hafta* that have been emerging in recent years (Krug 2000) are also represented in the child-directed corpus and presumably will be learned as such. Furthermore, while *do* is only rarely followed by a bare infinitive in the child-directed input data, all learners presumably will eventually hear a sufficient amount thereby learning the emphatic use of *do* — which, according to the decision tree in Figure 1, should be classified as a modal. Modal may be a somewhat unconventional way to treat the emphatic *do*: After all, the absence of agreement is typically viewed as a defining characteristic of modals (e.g., Huddleston & Pullum 2002, 107), but the emphatic *do* does show agreement. We simply wish to note that at stake is not how linguists choose to define word classes, but how children sort words in their language into distributional classes and use them appropriately. In the same vein, the verb *need* is primarily used as a lexical verb, and presumably learned as such initially. But it is also a modal as we can find examples such as *you need take a nap* and *they needn't do that* in the child-directed English corpus. And it is possible to find agreement on the modal form of *need* (e.g., "a major university needs not only excel academically") in larger samples such as the COCA corpus (Davies 2008).

This brief foray into the multifaceted profile of words such as *do*, *have*, and *need* is to demonstrate that the category status of words is empirically determined: if distinct distributional profiles of a word are available, and appear frequently enough to be learned, it will be assigned as members of distinct syntactic categories. While their usage will give the impression of variability — for example, in child-directed English *need* is used as a modal in fewer than 1% of all instances — it reflects a probabilistic ensemble of discrete representations and processes typical in the variationist approach to language (e.g., Sankoff 1988, Kroch 1989). There is no sense in which *need* is 99% of a lexical verb. This point is worth making because the phenomenon of grammaticalization has often been described as a gradual one (Hopper & Traugott 2003). Indeed, our own analysis of the modal verbs in English, especially Figure 2 in Section 6, clearly shows that in the period of Middle English, items such as *connen* ('can') alternated between modal and lexical usage with the modal form steadily gaining the upper hand. Under the current proposal, the gradualness of grammaticalization, and its extension on an item-by-item basis, is expected as children construct syntactic categories based on words and their distributions, which also develop gradually and on an item-by-item basis.

While a full treatment of syntactic categorization will take us too far afield (see Liang et al. 2022 for a preliminary computational implementation), we hope to have made the case that the major syntactic classes of English can be formed distributionally on a developmentally appropriate model of vocabulary acquisition and formal learning. We have so far focused on the immediately adjacent context for the classification of verbs (e.g., to right of the pronoun *I*, to the left of negation and bare infinitive). There may be other cues to further bolster the category distinction but we have refrained from making use of them in our discussion. For example, in PDE, only functional but not lexical verbs invert with subject pronouns in question formation. But to make use of this property for syntactic categorization, the child learner must have acquired the inversion process, which at least requires some preliminary knowledge of the basic syntactic structures such as the kernel – which in turn requires some knowledge of syntactic categories: a potential circularity. There is no doubt that children learn the inversion process very early, and indeed well before age 2 (Brown 1973), it may not be early enough to be developmentally available for the formation of major syntactic categories. For these reasons, we have only appealed to strictly local cues with direct evidence from infant distributional learning, including our treatment of modals in the history of English to be discussed in later sections.

The clear distributional distinctions across verb classes are strongly consistent with children's acquisition of syntactic categories more generally, and their acquisition of modal verbs in particular. In addition to the findings reviewed earlier which primarily build on behavioral measures in experimental conditions, children's language production also shows extremely early knowledge of the major syntactic categories. Misuse of syntactic categories is vanishingly rare (Valian 1986), and the use of auxiliary verbs in child language is early and almost completely error-free (Brown 1973). Observational studies of early child English confirm the status of functional verbs as distinct from lexical verbs. Peters & Menn (1993), for example, find clear evidence of a structural slot, which these authors dubbed *proto-modal*, that appears after the subject but before the main verb, even at a stage of child speech that still contains many reduced phonological forms of these elements. Finally, we examined the text transcript of modal verbs *can* and *will* in a 2-million-word corpus of child English (MacWhinney 2000): not a single error is found out of over 40,000 instances.

Nevertheless, children's semantic and pragmatic knowledge of modals lags far behind; see Papafragou (1998) for review. For example, Moore et al. (1990) find that three-year-old English-learning children show no understanding of the basic deontic and epistemic contrast in the usage of modals, which only begins to emerge gradually between age 4 and 6. The nature of such delay is open to question, as the function of modals lies in the intersection between language acquisition and conceptual development (Leahy & Carey 2020). These facts are in line with acquisition research (e.g., Naigles 2002) that has consistently shown children to prioritize formal properties of language even in the presence of salient semantic cues such as animacy and the count/mass (i.e., enumerability) distinction (Levy 1983, Gordon 1985). For our purposes, it suffices to note that modals are learnable as a formal syntactic category on simple distributional evidence. The psychological theory of learning, including the TP, will be explored as the mechanism that underlies the phenomenon of grammaticalization in Section 5.

4 the comparability of child data and historical data

In the previous section we have laid out our approach of the acquisition and learnability of the major classes of verbs in Present-Day English including modals. Under this approach, grammaticalization is defined as the construction of syntactic categories on the basis of distributional patterns in the data children are exposed to, and the TP, a formal learning principle of category formation. In the remainder of the article, we will add the diachronic perspective and demonstrate how our new approach can be applied to historical data to explain how children acquired this set of verbs in Middle English and Early Modern English times. More precisely, our aim is to examine whether the local contexts of negation and the selection of bare infinitives, which we identified as the distributional signatures "_not/n't" and "_bare infinitive" for PDE, also consistently identified the modals as a distinctive class in Middle English (ME) and Early Modern English (EModE) (and Modern British English, ModBE). We focused on these historical periods because in ME and EModE a number of major morphosyntactic and syntactic changes happened that changed the language to a considerable degree. So the question is how children confronted with input that was undergoing change were also able to identify modals from other types of verbs using distributional learning and the TP. We hypothesize that based on the way early child language acquisition proceeds children would go for local contexts first and find clear distributional distinctions in the same way as in PDE. This implies that our approach seeks to identify stages in these historical periods at which children first used the set of verbs that became modals in a novel way, i.e. with modals' distributions. Put differently, it is not the diffusion of that innovation that we will model.

Before we develop our diachronic approach to distributional, formal learning in Section 5, we will first discuss why we assume that child data is comparable to historical data at all.

Child language acquisition has been acknowledged by many scholars as a driving force of change (e.g. Paul 1880, Andersen 1973, Lightfoot 1979, Yang 2002). As a consequence of linguistic uniformitarianism (Labov 1972, Walkden 2019), in former times children must have acquired language in a similar way as they do today. So to examine the relationship between acquisition and change we need a method that allows us to compare the lexicons of children back then and today, because that is what they would have started out with. With his method of frequency trimming, Kodner (2019) has methodologically established the comparability of historical data and child data under the assumptions that (i) grammar learning takes place via type frequencies in the input, (ii) the token frequencies of types correlate with the age at which children acquire their lexicons and (iii) the main part of grammar acquisition is undertaken on the basis of a small early lexicon, i.e. few, mainly highly frequent types (also supporting the TP, see the previous section). Using his method, Kodner found substantial similarities, i.e. a semantic overlap of types, between lexicons derived from child directed speech and historical adult corpora that were trimmed by token frequency to approximate child lexicon sizes. These results provide the foundation for modelling language acquisition over historical data (for studies applying his approach see e.g. Sneller et al. 2019, Trips & Rainsford 2022 Kodner 2023, and Yang 2024).

We have used Kodner's method to demonstrate that the corpora we use here are comparable as well. Table 1 shows the 20 most frequent verbs in the spoken part of the BNC (serving as reference point), CHILDES and the Penn ME combined corpora. Among the 10/15 most frequent verbs in all corpora are the auxiliaries *have*, *be* and *do* as well as the modals *will*, *can* and *shall*. Among the 20 most frequent verbs in all corpora are verbs of motion (*come*, *go*), verbs of saying (*say*, *tell*, *speak*), verbs of cognition (*see*, *look*, *know*, *think*) and verbs of transfer of possession (*give*, *take*). These findings corroborate Kodner's method as they show a substantial semantic overlap of the most frequent verb types a child is exposed to in all three corpora. Furthermore, they support the approach of distributional learning and category formation using the

TP that we will apply in Section 6 to historical data.

The historical data were all extracted from the Penn Parsed Historical corpora of English. For ME we used the three verb lemmatised corpora available for this period (Percillier & Trips 2020): The *Penn-Helsinki Parsed Corpus of Middle English* (PPCME2, Kroch & Taylor 2000, release 4), the *Parsed Linguistic Atlas of Early Middle English* (PLAEME, Truswell et al. 2018), and the *Parsed Corpus of Middle English Poetry* (PCMEP, Zimmermann 2014). We refer to them as the combined ME corpora and use the periods as defined in the corpora: M1 (1150-1250), M2 (1250-1350), M3 (1350-1420), M4 (1420-1500). For EModE we used the two corpora available for this period: The *Penn-Helsinki Parsed Corpus of Early Modern English* (PPCEME Kroch et al. 2004, release 3), and the verb lemmatised version of the *Parsed Corpus of Early English Correspondence* (PCEEC, Taylor et al. 2006, Taylor et al. 2006). The periods defined in these corpora are E1 (1500-1569), E2 (1570-1639), E3 (1640-1720). To get an impression of how the distributional profiles further developed beyond EModE, we also included data for Modern British English (ModBE with periods of MB1 (1720-1799) and MB2 (1800-1897)) from the *Penn Parsed Corpus of Modern British English* (Kroch et al. 2016). For all corpora we used *CorpusSearch* to define our queries as well as the coding function[1]. Further, we produced frequency lists for distributional profiles to replicate the approach we proposed for PDE.

BNC, spoken part			**CHILDES**			**ME combined corpora**		
rank	**lemma**	**freq**	**rank**	**lemma**	**freq**	**rank**	**lemma**	**freq**
1	be	589675	1	be	11305	1	ben ‘be’	57089
2	have	203372	2	do	10243	2	haven ‘have’	15071
3	do	171766	3	go	4808	3	shulen ‘shall’	13416
4	get	95491	4	do neg|not	3969	4	seien ‘say’	9487
5	go	71338	5	have	3534	5	mouen ‘may’	7967
6	say	63330	6	get	2973	6	don ‘do’	7868
7	know	61719	7	see	2301	7	comen ‘come’	6379
8	will	56883	8	know	2103	8	willen ‘will’	5921
9	think	52349	9	put	1951	9	maken ‘make’	4847
10	can	49920	10	think	1901	10	sen ‘see’	3810
11	would	46242	11	can	1732	11	taken ‘take’	3112
12	see	32843	12	come	1546	12	yeven ‘give’	2758
13	come	31685	13	say	1394	13	wenden ‘go’	2350
14	mean	26161	14	will	1314	14	witen ‘know’	2026
15	want	23950	15	want	1286	15	gon ‘go’	1996
16	take	20836	16	look	1245	16	tellen ‘tell’	1809
17	look	20496	17	make	1115	17	thinken ‘think’	1770
18	could	20148	18	take	1058	18	speken ‘speak’	1742
19	make	19656	19	like	1006	19	bringen ‘bring’	1599
20	put	18908	20	tell	952	20	leten ‘let’	1580

Table 1: Comparability of data from the BNC, CHILDES and Penn ME corpora in terms of verb frequency

[1] See the CorpusSearch Reference Manual, `https://www.ling.upenn.edu/mideng/csdocs/CSRef.htm`, 12.10.21

5 grammaticalization and the emergence of modals in the history of english

In this section we will provide a brief state of the art of papers that have dealt with the development of modals in historical linguistics with a focus on two accounts that have figured most prominently and are most relevant to our study. Further, we will discuss some recent studies of negation and, related to it, the loss of verb movement, in ME and EModE. This will serve as a basis for identifying the distributional profiles for learning modals in Section 6.

The development of modals in the history of English has been a long-standing and hotly debated topic in research on syntactic change, and unsurprisingly a substantial body of literature exists accounting for this development from different perspectives with different aims (see Standop 1957, Visser 1963,§§1483-1561, Lightfoot 1979, 1991, 2020, Plank 1984, Warner 1993, Roberts & Roussou 2003, Fischer 2007 and many more; for an overview see Gregersen 2020). Since Lightfoot's (1979) controversial claim that today's set of core modal verbs (*can/could*, *may/might*, *shall/should*, *will/would*, *must*) arose through radical recategorisation, i.e. a single change in the abstract system, many authors (e.g. Plank 1984), and especially Warner (1993), have convincingly shown that the development of this set of verbs is best analysed as a gradual syntactic change. Although the verbs that were to become modals exhibited "verblike" formal properties in Old English (Warner 1993,98-100), they also already had a number of distinct formal properties that set them apart from the rest of the verbs (e.g. taking the bare infinitive, see Warner 1993,ch. 6). Overall, there is agreement that the rise of the set of English modals is a clear case of grammaticalization in Meillet's (1912) sense, i.e. where content words become functional words in the course of time.

Lightfoot's 1979 monograph was the first that explicitly looked at syntactic change from an acquisitionally-driven perspective in the generative framework and sought to account for the rise of modals. Lightfoot (1991) and (2006) are couched in the Theory of Principles & Parameters framework (Chomsky 1981, 1995) and assume that children follow a learning path to identify cues—grammatical structures—in their ambient language (input, E-language) to set parameters. Concerning the development of modals, Lightfoot assumes that a radical recategorisation took place at the beginning of the 16th century whereby today's modal verbs emerged from lexical verbs and became auxiliaries that then occurred in Aux/INFL. A number of independent gradual changes paved the way for this radical grammatical change (see also Fischer 2007). Throughout the years, Lightfoot updated his approach, see e.g. Lightfoot (2020) where he completely dispenses with parameters and cues and assumes parsing as the main mechanism of acquisition. Although Lightfoot's work explicitly addresses the role that acquisition plays in language change and has impacted the field of theoretical historical linguistics more than any other author in this way, he does not offer a precise and predictive account of the acquisition process. But an account of learnability does exactly have to have these properties. In Sections 2 and 3 we have introduced a quantitatively precise learning model of how children extract grammatical rules from the distributional properties of their input. In section 6 we will develop our precise and predictive account of the acquisition of modals in the diachrony of English.

The other account that we will mention here in more detail is Warner's (1993) authoritative study of the history of English auxiliaries. To this day, it is the most careful empirical work that traces the development of auxiliaries in the history of English including modals. It serves as an excellent basis to identify the main formal distributional properties in historical data that the child had to discover to acquire the set of modals.

As noted above, Warner finds verblike formal properties of the Old English verbs that developed into today's modals. The formal properties of modals that they also exhibited are given in the following list (Warner 1993,135)[2]:

[2]A further formal criterion might have been the absence of the aspectual prefix *ge-* which was present with all other verbs; for a discussion see Gergel 2017.

a. Subcategorisation for the bare infinitive (not *to*-infinitive)

b. Preterite-present morphology

c. Restriction of some verbs to finite forms

d. Use of past-tense forms without reference to past time reference

e. The availability of negative forms in *n* in Old English and Middle English (*nillan* etc.)

f. Absence of examples in which the verb is the antecedent to pro-verbal *do*.

These properties set them apart from the general class of verbs and identified them as a subgroup.[3] Warner further notes that in the course of time, the set of modals became part of the class of auxiliaries which overall became a more coherent class. Note, however, that his list is the accumulation of identifying all modal properties found in the data reflecting adult language use. It is not very likely that children would have been confronted with all of them to the same degree in their first stages of acquisition. In fact, we have demonstrated in Section 3 that only few distributional properties—"_not/n't" and "_bare infinitive"—are *sufficient* for the child to acquire the set of modals in PDE[4]. We will replicate this procedure here to see whether for earlier stages of English these two distributional properties were equally *sufficient* for the ME and EModE child to acquire modal verbs. The notion of sufficiency is key here. What we certainly need to take into account is, however, that especially ME was a transitional period with a number of ongoing syntactic changes which had a quantitative reflex in the distributional properties of the data.

The most relevant changes for our study are changes in the negation system (e.g. Wallage 2017) and the loss of (some types of) verb movement (e.g. Haeberli & Ihsane 2020), So in contrast to the data for Present-Day English, we expect to find much more variation in the properties that children had to identify in ME and EModE to acquire the set of modals. We will therefore discuss the development of negation and the loss of verb movement in the remainder of this section before we will present the analysis of our data in Section 6. The discussion of both developments is mainly based on the two papers mentioned above, i.e. Wallage (2017) for negation, and Haeberli & Ihsane (2020) for the loss of verb movement. This discussion has a descriptive purpose only, theoretical implications are not relevant for our study.

Since Jespersen's (1917,4) observation that there is a "curious fluctuation" in the development of negation (in a variety of languages) from simple pre-verbal marker to discontinuous marker (pre-verbal and post-verbal marker) to the subsequent loss of the original pre-verbal marker (in some languages), the so-called Jespersen's Cycle (Dahl 1979) has received considerable attention in (generative) research (e.g. Willis et al. 2013, van Gelderen 2011, 2024, Wallage 2017, and many others). For Middle English, the three overlapping, yet formally distinct stages are *ne*, *ne ... not*, and *not*, which are illustrated with the examples in (1) to (3).

Stage 1 (c. 1150-1300): *ne* functions as the sole sentence negation marker (in inversion (a.) and non-inversion contexts (b.), and as free and contracted element (a.), (b.) and (c.))

(1) a. wið-uten hem **ne** cumst ðu ðar naure!
without them NEG come you there never
'Without them you will never get there!'
(VICES1,25.265)

[3]For a thorough discussion of these properties and possible borderline cases see Warner 1993,ch.6.2.

[4]Both properties are given in Warner's list as properties a. and e. (see the discussion of negation below).

b. Y **ne** pursued hym þat bacbiteþ priuelich hys neȝbur
I NEG pursued him that bacbites prively his neighbour
'I didn't follow him who secretely spoke ill of his neighbour.'
(EARLPS,121.5304)

c. In al denemark **nis** wimman So fayr so sche, bi seint iohan!
in all Denmark NEG-is woman so fair so she by Saint John
'In all Denmark there is no woman as fair as she is, by Saint John!'
(Havelok,47.1715.807)

Stage 2 (1150-1400): Sentence negation *not* occurs with *ne* (as free and contracted form (a.) and (b.)):

(2) a. Certes, he that is fool-large **ne** yeveth **nat** his catel, but he leseth his catel.
certainly he that is foolishly-generous NEG gives NEG his property but he loses his catel
'Certainly, he who is improvident doesn't give his property unless he loses it.'
(CTPARS,316.C1.1189)

b. He **nuste** **nouht**. þat he wes. boþe god and mon
He NEG-know not that he was both god and man
'He didn't know that he was both, God and man.'
(PassionLord,38.36.30)

Stage 3 (1350-1500): *not* alone marks sentence negation:

(3) Prowde men or women lufes **noght** stalworthly;
proud men or women love NEG vigorously
'Proud men and women don't love vigorously.'
(ROLLEP,112.847)

Focusing on the distribution of *ne* and *not*, we see that throughout ME, sentence negation stands in a special relation to the finite verb: as long as *ne* exists (stages 1 and 2), it occurs almost always left-adjacent to the finite verb, sometimes in contracted forms[5]. In almost all of these cases it is the auxiliaries *haven*, *ben* and the modals *shulen* and *willen* as well as *witen* that occur in contracted forms (see Warner 1993,ch.9 and Section 6.1). For children, this may have been a cue for the distinction between lexical verbs and auxiliaries. As soon as *not* gains sentence negation function (stage 3) and as long as lexical verbs could directly occur in front of *not* due to verb movement (see below), there was no clear cue for children for the distinction between auxiliaries (including modals) and lexical verbs. But as soon as lexical verbs had to occur to its right, contexts with *not* could serve that function. This is when another crucial development in ME and EModE times comes into play, i.e. the loss of verb movement.

There is general agreement that in late ME (1350-1500) verb-movement to the inflectional domain still occurred. Placement of adverbs and the sentence negation *not* to the right of the verb are taken to show that the finite verb has moved out of the VP (see (4) from Haeberli & Ihsane (2022,499), subperiods and dates were added by the authors.)

(4) a. þerfor I *aske* **now** mercy (VAdv)
Therefore I ask now mercy.
'Therefore, I now ask mercy.'
(M4, 1450, KEMPE,141.3272)

[5]There are rare cases of non-adjacency of *ne* and finite verb: *ne ic cume to heom nawiht*. (LAMBX1,31.388).

b. Bott I *sawe* **noght** synne. (VNeg)
But I saw not sin
'But I did not see sin.'
(M4, 1450, JULNOR, 60.289)

Haeberli & Ihsane (2022) assume that verb movement is strongly retained up to 1650, at which point it then declines until the 18th century. The order *V-not* is replaced by *do*-support and it starts to occur in the 16th century.[6]

How do we interpret these empirical observations in our approach? According to the dates Haeberli & Ihsane (2022) give for the loss of *V-not* orders, only after 1650 would children have had evidence for this distributional signature to distinguish auxiliaries from lexical verbs. We will discuss the robustness of this signature in Section 6.1.

Considering further developments in EModE times relevant to our study, we refer to Warner (1993), who states that in Late ME and at the beginning of the EModE period a number of changes took place that sharpened the distinction between modal verbs and lexical verbs (ch.9), e.g. a further restriction in distribution of the bare infinitive (his change (C)). By the 16th century the distribution of the *to*-infinitive and the bare infinitive looks like the situation in PDE, i.e. according to Warner there is a clear opposition 'modal+ plain infinitive' and 'non-modal + *to*-infinitive' (p. 204). Furthermore, Warner gives a number of new developments, and critically for our study, the appearance of negatives in *-n't* that led to an internal coherency of auxiliaries and their distintcness from lexical verbs (Warner 1993,ch.9.2). According to the author, the contracted negative *-n't* is first found in the second half of the 17th century. This means that at this point children would have had further evidence for the difference between auxiliaries and lexical verbs. We will provide some numbers in Section 6.2 to discuss this point in more detail.

6 the distributional profile of modals in middle and early modern english

In this section we will take the perspective of a child acquiring the grammar of ME and EModE, and more precisely, the set of modals. It is crucial to understand the change in perspective: our interest is *not* to discuss the development of modals over time by quantifying language use of adult speakers/writers, which is what historical linguists usually do. Instead, we assume that ME and EModE data can be used to explain how children discovered the category of modals based on the assumptions that both types of data are comparable (see Section 4), and that they use distributional learning and the TP (see Section 2 and 3). For consistency and ease of comparison with the categorization of modals in PDE (Section 3), we focus on the ten most frequent verbal items in Table 1 and study their distributional profile in ME and EModE. More precisely, we show that modals could be identified as a distinct syntactic category from other verbs in these earlier periods of English: their distributional signatures in fact persist into PDE — namely, "_not/n't" and "_bare infinitive", as studied in Section 3 — with only minor variation that reflects independent developments in the history of the language.

6.1 Middle English

Since we do not have as much data for ME as we do for PDE child data, we used the combined ME corpora (ca 1.6 mio words) and replicated the method described above; however, to gain more data we took all sentences that start with a subject pronoun (Spron) and identified the subject-pronoun followers directly

[6] For further details about the assumption that verb movement is not a single event, see Haeberli & Ihsane 2022.

right-adjacent to the pronoun.[7] The majority of these elements are verbs[8]. We have seen in Table 1 that among the ten most frequent verbs are *ben* 'be', *haven* 'have', *shulen* 'shall', *seien* 'say', *mouen* 'may', *don* 'do', *comen* 'come', *willen* 'will', *maken* 'make', and *sen* 'see'. In line with what we said in Section 2.2 the profile that children would have been confronted with is

subject pronoun–*ben/have/mouen/shulen/willen/comen/don/maken/seien/sen*

Crucially, proclitic sentence negation *ne*[9] frequently occurs with *ben* (e.g. *nis nes*, *nam*, *nare*), *have* (e.g. *nefst*, *nafst*, *nabbe*), *mouen* (*nemei*), *shulen* (*neschal*), and *willen* (e.g. *nele*, *nule*, *nolde*; see again examples (1) c. and (2) b.). According to Warner (1993,151) "The verbs involved are either preterite-present or members of a potential auxiliary group, [...] But among verbs the property of possessing negative forms overlaps substantially with a potential auxiliary group, and may have helped contribute to its definition." So when negation is contracted with the finite verb, only a subset of the verbs take part, i.e. auxiliaries including modals. This is corroborated by our data. At this point, and when children still have a small vocabulary, they have evidence then that the verbal class is not homogeneous because proclitic negation splits lexical verbs from auxiliaries. The TP clearly predicts this: five among ten in a set of items is enough to trigger a subdivision since the TP threshold θ_{10} is 4. As Table 2 shows, this is a likely scenario for subperiods M1 and M2 after which point sentence negation *ne* was gradually lost. As long as it exists and contracted forms frequently occur (compare the token frequencies in the table across periods), children can make a subdivision on the basis of this signature. None of the most frequent lexical verbs that we identified in Table 1 (*comen*, *don*, *maken*, *seien*, *sen*), nor actually any lexical verb, occurs with contracted *ne*. From these distributions (0/164 in M1 and 0/160 in M2 for VB) children would have been able to make a clear-cut distinction between lexical verbs and auxiliaries. This clearly is a difference to PDE where children have robust evidence throughout that "_not/n't" is a distributional profile that allows them to subdivide auxiliaries from lexical verbs.

Table 2 also displays the further development of negation in ME, i.e. the rise of the sentence negation *not*. Comparing the token frequencies for *V-not* and *not-V* the former pattern is more frequent across all verb types, with the highest frequencies for modal verbs. Note that the numbers in the table refer to the 10 most frequent lexical verbs, modals, and the two auxiliaries *haven* and *ben* from Table 1. The numbers for the five most frequent lexical verbs are not very high, and more crucially, they occur left-adjacent and right-adjacent to *not* due to the ongoing syntactic changes discussed in Section 5 in detail. The two examples in (5) illustrate that at that time lexical verbs could occur left-adjacent and right-adjacent to *not*.

(5) a. A thyng I rede þe, þat þou **forgete noght** þis name Jhesu
a thing I advise you that you forget not this name Jesus
'One thing that I advise you is that you don't forget this name, Jesus'
(M4, 1450, ROLLEP,81.325)

[7] With 27% of the sentences in the ME corpora showing Spron-V-XP order, we assume it is a frequent word order; compare 8% of XP-V-S order.

[8] Total of Sprons: 58,176, PRO-VB-X (21,153), PRO-MD-X (8,119), PRO-BE-X (8,821), PRO-HAVE-X (3,489), PRO-NEG-X (2514), PRO-ADV-X (2,362), PRO-DO-X (1,102).

[9] For discussion of the status of *ne* as sentence negator see Wallage 2017.

	M1 (1150-1250)	M2 (1250-1350)	M3 (1350-1420)	M4 (1420-1500)
ne-V (free)				
ne-BE	146	179	29	9
ne-HAVE	61	118	29	0
ne-VB	164	160	14	2
ne-MD	658	495	57	15
ne-V (contr)				
ne-BE	598	267	105	12
ne-HAVE	166	45	14	0
ne-VB	0	0	0	0
ne-MD	264	104	55	4
V-not				
BE-not	18	194	491	230
HAVE-not	4	29	84	42
VB-not	8	37	85	25
MD-not	61	359	965	570
not-V				
not-BE	6	60	101	51
not-HAVE	0	2	12	9
not-VB	5	39	97	45
not-MD	0	0	0	0

Table 2: Negation patterns in the combined ME corpora with the 10 verbs from Table 1 (adjacency of verb and negation)

b. Many euydens haue I mad in my book Concordia þat Seint Ruffus **not be-gan** þis ordr, ...
many evidence have I made in my book Concordia that Saint Rufus not began this order
'I have gathered many pieces of evidence that Saint Rufus did not found this order ...'
(M4, 1450, CAPSER,147.42)

At this point, the distribution of verb and *not* was not a cue for children to build distinct categories for "auxiliaries" and "lexical verbs" as it is the case in PDE. Only with the loss of verb movement starting around 1650 did *not* become a cue to split lexical verbs from auxiliaries (see Section 6.2). Since by M4 contracted forms with proclitic *ne* were no longer available none of the sentence negations (*ne*, *not*) could be used by children to subdivide.

But what about the other distributional profile—"_bare infinitive"—that we identified for PDE? If we consider the next successor of the most frequent subject pronoun followers—subject pronoun-verb/modal/be/have-X—for all of them we find different grammatical forms of verbs: for the lexical verbs it is the *to*-infinitive (*comen to speken*), and for modals it is the bare infinitive (*will finden*). Note that in order to recognize the distribution of "bare infinitive", the child must be able to recognize the infinitive form in their language. In PDE, the infinitive has no morphological marking: as reviewed in Section 2, even prelinguistic infants can reliably segment off inflectional endings such as *-ing* and *-ed* to discover the stem and thus the infinitive. In older varieties of English such as ME, however, the infinitive is morphologically marked (e.g., *com**en*** and

finden above), similar to present day languages such as French (*ven**ir*** 'come', *trouv**er*** 'find') and German (*komm**en*** 'come', ***finden*** 'find'). While we cannot travel back in time to study how language acquisition took place in former periods of English, there is no doubt, based on numerous acquisition studies, that very young children would have clear knowledge of the finite/infinite distinction; see Wexler 1994 for a summary. For example, even at 20th month, the very beginning of multi-word utterances, French-learning children almost always use a finite verb before negation and adverbs, and an infinitive after (Déprez & Pierce 1993). We also note in passing that previous research has often attributed such cases of very early acquisition as evidence for innate structures and parameters (Wexler 1998, Yang 2002), but these findings are clearly compatible with a distributional learning advocated and reviewed here. More pertinently, we can safely assume that child learners of ME had full access to the form and distribution of infinitives in their construction of modals as a syntactic category.

Table 3 shows that the occurrence of the bare infinitive with modal verbs, i.e. the profile "_bare infinitive", was very robust throughout the ME period and a distinct property that set them apart within the class of verbs (clear-cut distinction in token frequencies between lexical verbs and modals).[10]

	M1 (1150-1250)	**M2 (1250-1350)**	**M3 (1350-1420)**	**M4 (1420-1500)**
bare inf.				
VB	0	0	0	0
HAVE	0	0	0	0
BE	17	6	7	0
modal	2073	3048	3657	3185
to-inf.				
VB	122	120	154	166
HAVE	6	8	13	9
BE	87	220	556	113
modal	0	0	0	0

Table 3: Bare infinitive vs. *to*-infinitive in the combined ME corpora with the 10 verbs from Table 1

Overall, the two distributional signatures identified for PDE could not have been used by ME children in the same way to acquire the set of modals. In M1 and M2 they were likely to use forms with proclitic *ne* to distinguish auxiliaries from lexical verbs but they could not have used "_not/n't" to do so. In contrast, the profile "_bare infinitive" was very robust in ME times for children to make a distinction between modal verbs and non-modals verbs. Figure 2 shows how four modals under scrutiny (*connen*, *mouen*, *shulen*, *willen*) alternated between modal and lexical usage with the modal form steadily and gradually gaining the upper hand. In effect, we classified these verbs purely on the distributional criterion that a child learner at the time would be able to establish. Clearly, as the figure also shows, this development happened on an item-by-item basis: all these items alternated between modal and lexical usage, and they progressed over time at different rates. As discussed in Section 3, such mixtures can be observed in PDE as well: while they are almost exclusively used as modals — *shall* has virtually disappeared from child-directed American English — an occasionally lexical usage can still be found and learned (e.g., *The captain willed the team to victory*). The gradual development of modals over time is a necessary consequence of lexical acquisition

[10] *Ouen* 'ought' occurs both with the bare and the *to*-infinitive until the *to*-infinitive is finally generalized. Warner's explanation is that, in contrast to the core modals, *ought* could always be used to express past time reference (Warner 1993,204).

and syntactic categorization.

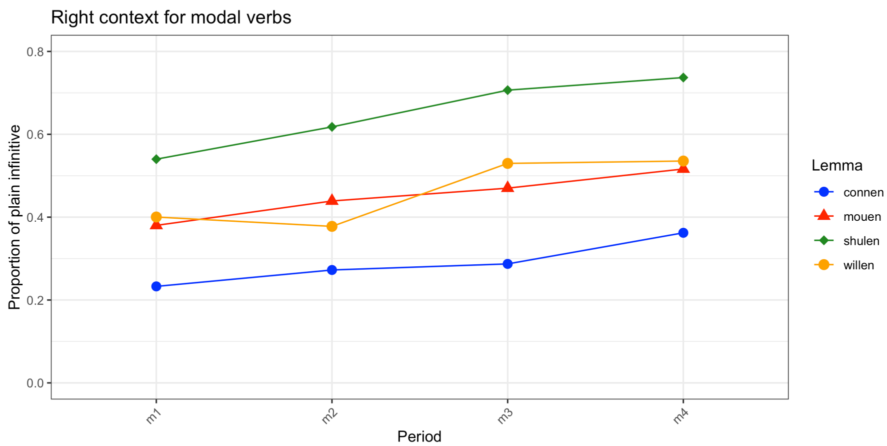

Figure 2: The development of the modals *connen*, *mouen*, *shulen*, *willen* on the basis of proportions of the right context

6.2 Early Modern English

In this section we discuss further developments of the two distributional profiles identified for ME. Let us begin with negation by taking a look at Table 4 which exhibits the distribution of *not* in Stage 3 of Jespersen's cycle.

First, by EModE times the negation *ne* has completely dropped out of the English language, so data for contracted *ne* and auxiliaries is not longer available. Second, although the distribution of *V-not* still includes all verb types there is a considerable difference in token frequency between the lexical verbs, on the one hand, and the auxiliaries and modals under scrutiny, on the other hand: by EModE numbers for *VB-not* clearly have decreased. A comparison of the numbers for *VB-not* (22) and *not-VB* (150) reflects the loss of verb movement. At this point children are likely to have started to use the distributional profile of "_not" to build the category of lexical verbs. This trend continues, as expected, in ModBE (compare *VB-not* (15) and *not-VB* (186)).

Another development that we briefly addressed in Section 5 related to negation is the development of *-n't* which, according to Warner, made the class of auxiliaries more coherent and, at the same time, the class of lexical verbs distinct from auxiliaries (ch. 9.2). Warner states that *-n't* is first found in the second half of the 17th century, and Brainerd (1989,181) assumes that the "first concrete evidence" began to appear before 1630. We will scrutinize this in our ME and EModE corpora now.

In M3 and M4, we start to find the precursors of contracted negatives in *-n't* like *shallnot*, *willnot* but only in small numbers, and a large number of *cannot* (see Table 5).

	EModE (1500-1720)	ModBE (1720-1897)
V-not		
BE-not	5094	1373
HAVE-not	2490	448
VB-not	22	15
MD-not	9202	1352
not-V		
not-BE	964	246
not-HAVE	270	64
not-VB	150	186

Table 4: Distributional profiles with *not* and adjacent verb with the 10 verbs from Table 1 in EModE and ModB

	M3	M4	EModE	ModBE
***not* contraction**				
cannot	4	66	1335	699
shallnot	0	4	5	0
willnot	0	16	0	0

Table 5: *not* contraction in ME, EModE and ModBE

From EModE onwards we see the rise of *-n't* negatives with auxiliaries like *don't/didn't*, *can't*, and *won't*, see Table 6.[11] These forms are a clear marker of auxiliaryhood, so together with the loss of verb movement and with that an increase in the distributional profile of "_not/n't", children were likely to form the classes of auxiliaries and lexical verbs just like in PDE.

	EModE	ModBE
n't contraction		
don't/didn't	107	412
can't	40	127
won't	27	67

Table 6: *n't* contraction in ME, EModE and ModBE

Concerning the distribution of the *to*-infinitive and the bare infinitive, we have shown that it has always been quite robust, and by the 16th century looked like the situation in PDE. Children at that time could have used the distributional signature "_bare infinitive" to split modal from non-modals. The loss of the residues of verbal inflection (for the infinitive and the participles) strengthened this signature.

[11]Few examples are found of *shan't*, *mayen't*, *haven't* and *ben't* not given in the table.

Summarising our findings, we have seen that throughout the EModE period quantitative changes in the distributional profiles "_not/n't" and "_bare infinitive" gradually developed towards the situation that children find today when they acquire the set of modals. Our data showed that children were able to make a split between lexical verbs and auxiliaries by the end of the 17th century using the distributional profile "_not/n't". The rise of *-n't* negatives supported this split. The distributional profile "_bare infinitive" had already been quite robust in ME, so children acquiring modals in EModE were also able to use it to subdivide modal and non-modal auxiliaries. As the vocabulary of children increased they would have looked out for further cues to build these categories, for example, the ones that are listed by Warner. Still, in late EME the two distributional profiles under scrutiny were *sufficient* for children to acquire the major sets of verbs, and with that the class of modals. Table 7 summarises our results.

	Distributional profiles	
	"_not/n't"	"_bare infinitive"
early Middle English	- ("n+aux")	+
late Middle English	-	+
early Early Modern English	-	+
late Early Modern English	+	+
Present-Day English	+	+

Table 7: Distributional profiles in ME, EModE and PDE for modals

7 conclusions

This paper discussed the grammaticalization of modal verbs in child language acquisition in the diachrony of English. We described a new distributional approach that invoked the Tolerance Principle, and its recursive application, as a formal principle to syntactic category formation. When a set of items is *sufficiently* homogeneous with respect to its distributional properties, it can be regarded as a distinct syntactic category. Sufficiency is precisely defined by the TP threshold. If this is not the case, then subdivision is necessary and additional syntactic categories may be created. We found that this was the case when children acquire the major verb categories of Present-Day English and that two distributional signatures—"_not/n't" and "_bare infinitive"—are sufficient to define lexical verbs, auxiliaries and modals. We then demonstrated, with the use of corpus data, that children in ME and EME times could identify modals as a distinct syntactic category with essentially the same distributional signatures that also reflect the independent changes in the history of the language. In particular, we found that the distributional signature "_bare infinitive" was always available for children to build the class of modals. However, since negation underwent a number of changes in ME and EME times the distributional signature "_not/n't" could have only been used after 1650 when movement of lexical verbs was clearly lost. But in early ME times proclitic negation ("n+aux") would have been a distributional signature for children to build the category auxiliary (including both *have*, *be* and the modal verbs). Overall, we presented an approach that sheds new light on the grammaticalization of modal verbs, i.e. how children responded to changes in distributional data over time.

It is instructive to consider, in light of our approach, the development of modals in English in comparison to other languages especially its Germanic cognates. As is well known, the "core" modal verbs in English belong to the class of preterite-present verbs shared across the Germanic family. But membership in this class is clearly not necessary for the development of modals, as seen in the case of *need*, *ought*, etc., it is evidently not sufficient either as only English modals developed into a strongly distinctive class. At the same time, other Germanic languages, indeed all languages to our knowledge, have ways to express modality. The semantic and pragmatic functions of modality, therefore, cannot be the reason for the development of English modals as a distinct syntactic category. Nor are these functions necessary: As we have demonstrated in the present paper, the formal distributional properties are already *sufficient*, and children in any case acquire the semantic aspects of modality much later than the formal syntactic properties (see e.g. Papafragou 1998, Cournane 2019).

The perspective of distributional learning invites us to focus on the formal evidence for the modal verbs in English and those in related languages. While the ability to take a bare infinitive has been widely noted as a key property of modal verbs in all Germanic languages (e.g., Abraham 1990, Reis 2001, Mortelmans et al. 2009), English modals stand out more. In German, for example, auxiliary verbs denoting durative aspect, futurity or expressing the subjunctive (e.g., *bleiben* 'stay', *wird* 'will', *würde* 'would') can also take bare infinitives (e.g., Engel 1980, Maché 2019, 21). In Dutch, the bare infinitive can serve as the complement for aspectual auxiliaries (*blijven* 'stay' and *gaan* 'go'); see IJbema (2002, Chapter 2). In Norwegian, bare infinitives appear not only with modals but also auxiliaries such as the perfect (Eide 2005). But the strongest, and most unique, distributional evidence for English modals comes from the placement of negation. After all, English is the only language in the family that lost the movement of lexical verbs in the matrix sentence completely, which leaves the structurally high placement of functional verbs (modals and auxiliaries) — above negation — as the key distinguishing feature. From a distributional learning perspective, it is these independent changes in the morphosyntax of English that led to the emergence of a salient formal category of modals.

More generally, our approach to grammaticalization is part of a research program that returns to the foundation of generative grammar (Chomsky 1955), a discovery procedure that aims to distributionally derive all the substantive components of language from a corpus — child-directed input in the case of language acquisition. It does not assume any pre-determined inventory of syntactic categories or parameters, which have been the basic explanatory tools for language change in the generative tradition (Lightfoot 1979, Roberts & Roussou 2003). As we have illustrated, the structural properties of modals (e.g., _neg, _bare infinitive) are not inherent to these categories, and are certainly not innate or universal so as to cause ("trigger") their acquisition and change (Lightfoot 1991). Rather, they are *discovered* by the child learner from very simple data on distributional criteria: modals as a class are the psychological *consequence* of such distributional learning discoveries. In some ways, the TP can be viewed as a concrete proposal of the "Transparency Principle" that Lightfoot (1979) appeals to as an independent measure of grammatical opacity and exceptionality but never makes mechanistically or quantitatively precise. Along with these and many previous scholars, we embrace the centrality of language acquisition in the explanation of language change.

Our approach gains support from the simple fact that the distributional signatures for modals, which are motivated solely from the perspective of child language acquisition and formal principles of learning, converge with the observations from historical scholarship (Warner 1993). To the extent that syntactic categories can be defined directly from the language data, there would be no need for any independent mechanism of grammaticalization (Harris & Campbell 1995).

As we conclude, it is worth reiterating that our theory is about how syntactic categories are formed and may change in response to the changes in the language learning data. Why and how the data itself changes is, of course, part of the historical phenomenon but an independent matter that may or may not have tractable answers. Some of the changes in data such as the loss of verb movement may indeed receive structural explanations. Other changes in the data, however, are due to the so-called creative aspect of language use

(Chomsky 1966/2009): “the diversity of human behavior, its appropriateness to new situations, and man’s capacity to innovate” (p61). For example, someone in North America presumably decided to drop the auxiliary *be* so *gonna* became a standalone modal verb sometime before the year 1904, the first record of attestation according to the OED. Sometime before that, *gonna* appeared as the phonological reduction of *going to*. And that usage caught on. We are in no position to offer why, how, or when *gonna* became a modal: we do specify a mechanistic process by which children analyze *gonna* as a member of a distinct class — and use it as such — if they hear it used with a bare infinitive during the formative stage of language acquisition. Such is the nature of any historical science, where we strive to delineate the contingent from the necessary.

There is nothing in our method that restricts its applicability to modal verbs or indeed grammaticalization. Its explanatory force in historical linguistics is no more or less than its effectiveness in accounting for child language acquisition. Through the detailed studies of the English modals in the past and present, we hope to have demonstrated the utility of distributional learning in understanding how grammars are formed and how they may change.

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